Efe Obukohwo

Full Stack Developer | Python Developer | Data Engineer

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Portfolio: efeobus.github.io/Efe-Obukohwo | LinkedIn: linkedin.com/in/efe-obukohwo | GitHub: github.com/EfeObus

Professional Summary

Full Stack Developer with 5+ years of experience architecting production platforms across web, cloud, and native mobile. Shipped full-stack applications (FastAPI, Flask, Node.js/Express, React) with 500+ API routes and 88.3% ML prediction accuracy, plus cross-platform mobile apps (Kotlin Multiplatform, Capacitor) running natively on iOS, Android, and Web from a single codebase. Experienced across AWS and Google Cloud Run, integrating both OpenAI and Vertex AI (Gemini) into production features, with engineering practices spanning CI/CD pipelines, JWT/biometric authentication architecture, and observability (Prometheus, Sentry). Demonstrated ability to lead development teams, optimize database systems (25% efficiency improvements), and deploy scalable infrastructure with 99.9% uptime.

Core Competencies

Python • JavaScript • TypeScript • Kotlin • HTML5 • CSS3 • SQL • Java • React • Zustand • Tailwind CSS • Bootstrap • Vite • Responsive Design • Flask • FastAPI • Node.js • Express.js • Ktor • RESTful APIs • JWT • Plaid API • Docker • Git • Kotlin Multiplatform • Compose Multiplatform • Capacitor (Android/iOS) • SQLDelight • Koin (DI) • Biometric Auth • Push Notifications (FCM/APNs) • Google Cloud Run • Cloud SQL • Secret Manager • AWS • GitHub Actions (CI/CD) • Firebase • Prometheus/Micrometer • Sentry • Scikit-learn • XGBoost • LightGBM • Random Forest • AutoML • OpenAI API • Gemini (Vertex AI) • Pandas • Tableau • Power BI • Excel • SPSS • Apache Spark • Apache Hive • PostgreSQL • MySQL • Oracle • SQLAlchemy • Sequelize • Drizzle ORM • Prisma • Airtable

Professional Experience

Full Stack Developer Intern | Zodiac Tech Toronto
Toronto, ON | May 2025 – August 2025

- Architected an AI-powered real estate analytics platform processing 49,551+ Canadian properties, achieving 88.3% ML prediction accuracy with an ensemble of XGBoost, LightGBM, and Random Forest
- Led full-stack development of a production Flask application with MySQL and a 20+ endpoint REST API, shipping interactive property mapping and investment analytics
- Implemented a 5-tier API security system with ML-powered XSS/CSRF detection and behavioral analysis, achieving a 100% attack-prevention rate in testing
- Deployed a Dockerized CI/CD pipeline achieving 99.9% uptime, <400ms API response times, and 85% test coverage while leading a team of 5 developers

Database Consultant | Humanitarian Outcomes
Remote | May 2023 – Dec 2025

- Resolved data-integrity issues across the GDHO Airtable and database while designing optimized schemas with the team lead, driving a 25% increase in operational efficiency
- Conducted trend analysis using SPSS and SQL, then visualized findings in Tableau and Power BI to drive data-informed decisions for humanitarian stakeholders worldwide
- Built and segmented 1,200+ partner and audience lists using Airtable and SQL, increasing campaign engagement by 30%
- Qualified 350+ leads from events and surveys using structured frameworks, converting 120 into high-priority partnership opportunities

Customer Service Representative | IGNITE Student Life, Humber College
Toronto, ON | Aug 2024 – May 2025

- Resolved billing issues and account discrepancies across a high volume of inbound student inquiries via phone, email, and live chat
- Advised students on IGNITE services, wellness resources, and campus events while consistently meeting call-resolution and satisfaction targets

Compliance Officer - AML/Fraud | United Bank for Africa (UBA)
Sapele, Nigeria | Nov 2019 – Aug 2022

- Monitored and analyzed daily transactions across a retail and corporate client portfolio, detecting and preventing money laundering, terrorist financing, and fraud
- Investigated 30+ system-generated alerts weekly using SQL queries and advanced Excel (VLOOKUP, Pivot Tables) to identify and escalate suspicious activity
- Conducted Customer Due Diligence (CDD) and Enhanced Due Diligence (EDD) for high-risk clients, ensuring full regulatory compliance
- Drafted Suspicious Transaction Reports (STRs) for the Nigerian Financial Intelligence Unit, contributing to a 10% increase in branch reporting accuracy

IT Consultant | Finecare Diagnostic Limited
Sapele, Nigeria | Oct 2020 – Oct 2021

- Devised and executed IT infrastructure solutions, including network setup and maintenance, boosting operational efficiency
- Achieved a 65% increase in accounting accuracy by introducing invoicing software for receipts generation
- Conducted market research and customer surveys to identify business expansion opportunities

IT Support Specialist | Delta State College of Education, Nigeria
Mosogar, Nigeria | Nov 2018 – Oct 2020

- Led the installation and configuration of 40 workstations, ensuring optimal performance and 100% user satisfaction
- Resolved technical issues for 65+ end-users promptly and accurately while strengthening antivirus and cybersecurity protocols across all systems
- Implemented a remote support system, streamlining registration and payment assistance for students

Education

Ontario Graduate Certificate, Information Technology Solutions
Humber College, Toronto, ON | Jan 2024 – Aug 2025
GPA: 86.5%

Ontario Graduate Certificate, Research Analyst
Humber College, Toronto, ON | May 2022 – Aug 2023
GPA: 85.6% | consecutive Dean's Honors List Award

Bachelor of Science, Computer Science
Delta State University, Abraka, Nigeria | Dec 2016 – Oct 2019
GPA: 3.75 in a 5-point grading system

National Certificate in Education, Computer Science and Physics
Delta State College Of Education, Mosogar, Nigeria | Dec 2013 – Nov 2016
GPA: 3.92 in a 5-point grading system

Portfolio Projects

TEKVWA Money (April 2026 – Present)

• TEKVWA Money is a full-stack personal finance tracking application solving financial leakage — the silent loss of money through small, unnoticed expenses. Built with React, Express.js, PostgreSQL, Zustand, and Tailwind CSS, and shipped as a website plus native Android and iOS apps (Capacitor) from a single codebase. Features real bank-account linking via Plaid (live in Production on web, Android, and iOS), Gemini/Vertex AI-narrated spending insights, automated PDF/CSV/XLSX bank statement import with self-learning auto-categorization across 24+ Canadian banks and credit unions, budget-driven spending alerts, biometric app lock (Face ID/fingerprint), push notifications, and exportable PDF/Excel/CSV reports. Includes a full RBAC system (Customer/Staff/Admin/Super Admin) with a Super Admin operations dashboard for audit logging, security history, and maintenance mode, backed by a real automated test suite and 16 tracked database migrations.

Live: https://tekvwamoney.com/ | GitHub: https://github.com/EfeObus/Tekvwa-Money

Technologies: React, Express.js, PostgreSQL, Zustand, Tailwind CSS, Capacitor (Android/iOS), Plaid API, Gemini (Vertex AI), Firebase Cloud Messaging

VwaTek Apply (Feb 2026 – Present)

• Professional AI career suite built with Kotlin Multiplatform, delivering native experiences for iOS, Android, and Web from a single codebase with Koin dependency injection and a clean data/domain/presentation architecture shared across all three targets. Powered by Gemini 2.0 Flash, it provides comprehensive job search optimization including ATS resume scoring with keyword gap analysis, AI-powered contextual resume rewriting using X-Y-Z impact framework, tailored cover letter generation, and full resume version control. Features tough AI recruiter mock interviews with STAR method coaching, Canadian salary intelligence with NOC code matching and negotiation strategies, and LinkedIn profile optimization. Built production-grade: encrypted on-device persistence (SQLDelight), offline-first support with automatic cross-device sync and conflict resolution, and observability via Firebase Crashlytics, Sentry, and Prometheus/Micrometer metrics on the Ktor backend. Built for the Canadian market with full PIPEDA compliance and Canadian data residency (Montreal), with subscription tiers (Free/Pro/Premium) via Stripe, Apple IAP, and Google Play Billing, plus enterprise SSO support (SAML, OIDC, Azure AD, Okta).

Live: https://vwatek-apply-production.up.railway.app | GitHub: https://github.com/EfeObus/VwaTek-Apply

Technologies: Kotlin Multiplatform, Compose Multiplatform, Gemini 2.0 Flash, Ktor, Koin, PostgreSQL, SQLDelight, Firebase, Sentry, Stripe, Railway

JTOV Network (Jan 2026 – Present)

• An advanced AI-powered mentorship platform that connects professionals through intelligent matching, skill development, and interactive networking tools. Features include AI-powered smart matching for mentors/mentees based on compatible skills and goals, real-time messaging and WebRTC video calling, gamification elements with achievements and leveling, AI-generated career roadmaps, structured learning paths, and an analytics dashboard for tracking progress. Includes an AI Learning Hub with personalized tutoring, skill gap analysis, and smart recommendations. Features a complete video learning platform with course management, interactive quizzes, and progress tracking. Built with comprehensive admin/staff dashboards, audit logging, and support ticket system. Implements enterprise security including CSRF protection, rate limiting, secure sessions, and Helmet security headers.

GitHub: https://github.com/EfeObus/JTOV-Network

Technologies: React, TypeScript, Node.js, Express.js, PostgreSQL, Drizzle ORM, WebSockets, WebRTC, OpenAI API, TailwindCSS, Framer Motion

TekVwarho ProAudit (Oct 2025 – Present)

• Nigeria's premier enterprise-grade tax compliance, payroll, and forensic audit platform built for the 2026 Nigerian Tax Reform. Automates complex statutory obligations—PAYE, VAT, CIT, WHT, pension, and NRS e-invoicing—while providing audit-ready records, immutable evidence, and regulator-friendly reports for FIRS and external auditors. Features include a 2026-compliant tax engine, full Nigerian payroll system with progressive PAYE bands, world-class forensic audit capabilities (Benford's Law fraud detection, Z-Score anomaly analysis, 3-way matching, hash-chain immutable ledger), and NDPA 2023 security compliance with AES-256-GCM encryption. Built with 500+ API routes, 80+ database models, and 400+ automated tests.

Live: https://ingenious-emotion-production-c3c3.up.railway.app/ | GitHub: https://github.com/EfeObus/TekVwarho-ProAudit

Technologies: Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic, HTMX, TailwindCSS, Celery, Redis, Cryptography

WonderWorld Learning Adventure (Dec 2025 – Present)

• Research-driven adaptive early-childhood learning platform for ages 2–8, blending cognitive science, early-years pedagogy, and game design. Features a multi-domain curriculum covering literacy (letter formation, phonemic awareness, reading fluency), numeracy (subitizing, counting, early arithmetic), and social-emotional learning. Uses Item Response Theory (Rasch Model) for real-time personalization maintaining ~75% success probability. Includes multi-sensory SVG stroke tracing, on-device offline speech recognition, narrative game worlds with mascot-guided progression, and a parent dashboard with progress tracking. COPPA and GDPR-K compliant with offline-first architecture for low-connectivity environments. Achieves 60 FPS across mobile and web.

GitHub: https://github.com/EfeObus/WonderWorld-Learning-Adventure

Technologies: Flutter, Dart, Flame Engine, Python, FastAPI, PostgreSQL, Redis, NumPy, Speech-to-Text, Adaptive Learning

TekVwa Life (Dec 2025 – Present)

• Secure digital autobiography and legacy-preservation platform that enables users to write, organize, and protect their life stories using a professionally structured 28-chapter framework. TekVwa Life combines narrative creation, life-event and achievement tracking, and post-death legacy access into a single privacy-first system. Built with enterprise-grade security, Canadian data residency, and professional PDF/DOCX export, the platform transforms personal history into a permanent, transferable digital asset for families, historians, and estate planning purposes.

Live: https://tekvwa-life-production.up.railway.app/ | GitHub: https://github.com/EfeObus/TekVwa-Life

Technologies: Node.js, Express.js, PostgreSQL (AWS RDS), Prisma ORM, React, Material UI, JWT Authentication

Tekvwa Analysis Platform (Jan 2025 – Present)

• Tekvwa Analysis (v4.4.0, rebranded from TekVwarho Analysis) is a comprehensive statistical analysis platform with AI-powered reporting, migrated from Railway to Google Cloud Run (Cloud SQL, Secret Manager, Artifact Registry) with AI generation moved from the OpenAI API to Vertex AI (Gemini) authenticated via Application Default Credentials, eliminating any API key to store or rotate. Built around five core capabilities: an Excel-like data-entry grid for direct spreadsheet input, a deterministic auto-analysis engine that profiles uploaded data and recommends the statistically appropriate test with a plain-language rationale, plain-language interpretation panels (headline, key findings, caveats, glossary) on every one of 17 analysis pages, interactive drag/resize dashboards with per-chart pinning that persist per user, and AI reports that embed real chart images alongside Gemini-narrated analysis with a deterministic fallback so reports are never empty. Hardened app-wide login enforcement across every page and API call, and debugged a real data-type-detection bug that crashed the upload pipeline on fully-populated numeric columns, reproduced and fixed against a live 536,000-row dataset.

Live: https://tekvwarho-analysis-551794668026.northamerica-northeast1.run.app/ | GitHub: https://github.com/EfeObus/TekVwarho-Analysis

Technologies: Flask, PostgreSQL, Google Cloud Run, Vertex AI (Gemini), Cloud SQL, Plotly.js, Bootstrap, Python

TK Reports - AI-Powered Data Analysis (Oct 2024)

• Advanced AI-powered data analytics platform that transforms datasets into comprehensive, professional-grade analytical reports. Built with Flask and OpenAI GPT-4 Turbo, it combines statistical analysis (mean, median, mode, standard deviation, skewness, kurtosis, outlier detection), machine learning insights, correlation analysis, and natural language generation to produce executive-ready reports in DOCX and PDF formats. Features research objective-driven analysis, 5 professional visualizations with detailed interpretations, handles up to 100MB file uploads, supports CSV/XLSX/XLS/TSV formats, and delivers 15+ page comprehensive reports with AI-generated executive summaries, key findings, insights, and actionable recommendations.

GitHub: https://github.com/EfeObus/TK-Reports

Technologies: Python, Flask, OpenAI GPT-4, Pandas, NumPy, SciPy, Matplotlib, Seaborn

Zodiac Real Estate AI (May 2025 – Aug 2025)

• Production-ready real estate analytics platform with 88.3% accuracy machine learning, combining real-time economic data for property valuations and market analysis across 49,551+ Canadian properties.

GitHub: https://github.com/EfeObus/NextProperty

Technologies: Python, Flask, MySQL, Scikit-learn, XGBoost, Docker, JavaScript

IGNITE Sleep Lounge Automation (Sep 2024 – May 2025)

• Digital check-in and waiver system for Humber College's IGNITE facilities, automating student registration, time tracking, and staff oversight with real-time functionality and admin dashboard.

GitHub: https://github.com/EfeObus/IGNITE-Sleep-Lounge-and-Zenzone

Technologies: Node.js, Express, PostgreSQL, JavaScript, JWT, HTML/CSS

Tekevwe LegalEagleAI (Oct 2024 – Present)

• Web application designed to streamline legal document management and analysis. Provides features such as document generation, editing, and summarization, leveraging AI to assist users with legal research, contract creation, and risk analysis. The platform includes dashboards for activity tracking, quick actions, and recent documents, as well as collaboration tools with an AI chat assistant for insights and support.

GitHub: https://github.com/EfeObus/LegalEagleAI

Technologies: React, TypeScript, Vite, Tailwind CSS, Node.js, PostgreSQL, OpenAI API, Drizzle ORM

TekVwarho Properties (Feb 2025 – Present)

• Nigerian property rental platform connecting landlords and tenants directly, eliminating middlemen to ensure transparency, security, and automation in rentals. Features ID verification, fraud detection, secure lease storage, and real-time communication. Landlords pay ₦10,000/year for unlimited listings and auto-generated legal agreements, while tenants pay ₦5,000/year for advanced search and agreement management.

Live: https://tekvwarho-properties-production.up.railway.app/ | GitHub: https://github.com/EfeObus/TekVwarho-Properties

Technologies: HTML5, CSS3, JavaScript, Node.js, Express.js, PostgreSQL, Paystack, WebSocket

Agho Hospital Management System (2025)

• Comprehensive desktop and web application to streamline hospital operations with role-based access control. Features patient management, appointment scheduling, doctor management, and user administration. Includes both web interface (port 5140) and console mode with security features like password hashing, SQL injection prevention, and role-based permissions for administrators, doctors, receptionists, and nurses.

GitHub: https://github.com/EfeObus/Agho-Hospital-Management

Technologies: Java, Java Swing, PostgreSQL, JDBC, SHA-256, Web Server

Online Retail SQL Analysis (Jul 2026)

• Designed a normalized PostgreSQL database (5 tables + 1 staging table, 3NF, indexed foreign keys) for a 500,000+ row UK e-commerce transaction dataset, then wrote 30 production-ready SQL queries covering customer segmentation (RFM, LTV, cohort retention), product performance and cross-sell analysis, temporal trends, and anomaly detection using window functions (RANK, LAG, SUM OVER, NTILE, PERCENTILE_CONT), CTEs, and statistical aggregation. Surfaced that £8.91M in revenue from 4,339 customers is 82% UK-concentrated, that the top 10% of customers drive 61% of revenue, and that sub-£5 budget products generate 71% of revenue despite being a volume rather than margin business. Went further than reporting results: audited the query outputs themselves and documented real data-quality findings, including a CURRENT_DATE bug that silently mislabeled 100% of customers as "Inactive" and a cohort query that only ever returned month zero — the kind of self-auditing that separates production analysis from a one-off script.

GitHub: https://github.com/EfeObus/Online-Retail-SQL-Analysis

Technologies: PostgreSQL, SQL, Window Functions, CTEs, RFM Segmentation, Cohort Analysis, Database Design

Nashville Housing Market Analysis (Jul 2026)

• Engineered a raw, unstructured municipal housing dataset of 56,636 rows into a verified, stakeholder-ready business intelligence solution using advanced Excel. De-duplicated 168 exact-duplicate sales records, designed defensive null-value routing that classified 30,619 unvalued records as "N/A" instead of zero to prevent market-tier skew, and engineered derived time-dimension and boolean flag columns. Built a dynamic formula engine combining VLOOKUP, XLOOKUP, and INDEX/MATCH for record retrieval with SUMIFS/COUNTIFS/AVERAGEIFS for localized statistics, and replaced standard aggregations with SUBTOTAL and SUMPRODUCT logic so every metric recalculates safely under live filters. Delivered a BI dashboard with 9 pivot tables, 5 KPI indicators, and a slicer-driven 2x3 matrix chart layout. Uncovered $18.50B in verified transaction value across 56,468 sales — a $327,581 average against a $205,838 median sale price, flagging significant high-value outlier skew — progressing from 11,338 sales ($2.78B) in 2013 to a 16,794-sale, $6.71B peak in 2015 (52% dollar-volume growth) before cooling to 14,055 sales ($4.23B) in 2016. Also surfaced ~80% capital concentration in Nashville proper and an 8.6% vacant-property sale rate — insights translated into geographic allocation, seasonal listing timing, and risk-monitoring recommendations for stakeholders.

GitHub: https://github.com/EfeObus/Nashville-Housing

Technologies: Microsoft Excel, VLOOKUP/XLOOKUP, INDEX/MATCH, Pivot Tables, SUMPRODUCT, Data Cleaning, Business Intelligence

Certifications

• Hands-On Introduction SQL - LinkedIn Learning (Dec 2023)
• NVivo 2018 Essential Training - LinkedIn Learning (Feb 2023)
• Intermediate SQL for Data Scientists - LinkedIn Learning (Dec 2022)
• TCPS 2: Core 2022 Certificate - Panel on Research and Ethics Canada (May 2022)
• Jobberman Soft-Skills Training - Jobberman (Feb 2022)

Key Achievements

• Achieved 88.3% ML prediction accuracy in real estate analytics platform processing 49,551+ properties
• Improved database system efficiency by 25% through optimization and error resolution strategies
• Deployed scalable infrastructure with 99.9% uptime and <400ms API response times
• Consecutive Dean's Honors List Award with 85.6% cumulative GPA in Research Analyst program
• Led installation and configuration of 40 workstations with 100% user satisfaction